

Mathematical Analysis of Digital MASH Delta-Sigma Modulators for Fractional-N Frequency Synthesizers

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Abstract—A mathematical analysis is performed to investigate the periodic behavior of digital MASH Delta-Sigma modulators (DSM). The analysis is performed on first, second and third order MASH digital DSMs with odd initial conditions on the first stage. We prove that the maximum sequence length for the first order modulator is 2^N and for the second order modulator is 2^{N+1} , where N is the word length of the state. In the case of first and second order modulators, the maximum sequence length can be achieved when odd digital inputs are applied. In the case of the third order modulator, the sequence length is equal to 2^{N+1} for all digital inputs.

I. INTRODUCTION

Delta-Sigma modulators were introduced in [1] and [2] to shape the quantization noise power of fractional-N frequency synthesizers. Linear noise shaping can be achieved if the highly nonlinear block model of the delta sigma modulator, the quantizer, can be modelled by an additive white noise source. It has been proven [3] that, even with a continuous-amplitude discrete-time model, inherited from oversampled delta sigma analog to digital converters, the modulator behavior can be periodic when certain dc inputs are applied and thus, spurious tones appear in the modulator's output noise power spectrum.

Gray showed mathematically [3] that for a first order modulator with dc inputs, the output spectrum consists of discrete spurs whose locations and amplitudes depend on the input value. This analysis was extended to higher order multistage noise shaping modulators in [4] and [5]. The results in [4] and [5] have proven mathematically that, in a higher order MASH modulator, the additive white noise source assumption for the quantizer error from the last stage is correct, when *irrational* dc inputs are employed.

These mathematical analyses of modulators were performed with the assumption of zero initial condition on the accumulators. For the case of irrational dc inputs, the asymptotic behavior of the quantizer error is not affected by the initial conditions [3], [4], [5] and [6]. Kozak and Kale [6] provide an exact analysis for higher order MASH $\Delta\Sigma$ modulators with rational dc inputs and non-zero initial conditions. Their approach effectively shows that an irrational initial condition imposed on the first accumulator guarantees a tone-free output

spectrum for third and higher order MASH modulators when driven by rational dc inputs.

In attempting to extend their theoretical results to the case of *rational* initial conditions (digital DSMs), Kozak and Kale suggest that if the internal digital word length of a fixed-point digital DSM is relatively high, irrational initial conditions can be implemented using odd numbers, although, their mathematical analysis does not predict the periodicity of digital DSMs.

Borkowski *et al.* [7] have used the results of [6], but have considered digital $\Delta\Sigma$ modulators as finite state machines (FSM). They performed extensive simulations on the various orders of MASH and error feedback modulators in order to extract an empirical design methodology for digital DSMs. Their empirical results for second and third order MASH digital DSMs are summarized in Table I, where I_1 is the initial condition of the first stage.

TABLE I
SEQUENCE LENGTHS FOR SECOND AND THIRD ORDER MASH DSMs
WITH WORD LENGTH N [7]

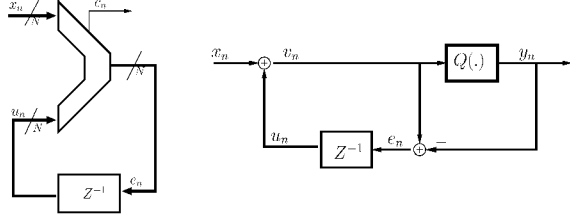
Modulator Order m	Guaranteed Sequence Length	Maximum Sequence Length	Initial Condition
2	2^{N-1}	2^{N+1}	I_1 odd
3	2^{N+1}	2^{N+1}	I_1 odd

In this paper, we prove these results. We calculate rigorously the sequence length of digital MASH DSMs. The analysis is performed on first, second and third order MASH DSMs with odd initial conditions in the first stage and verifies the empirical observations of [7].

II. FIRST ORDER MODULATORS

In this study, we derive a closed form expression for the binary quantizer error of a first order digital modulator and we calculate its sequence length (period). The accumulator shown in Fig. 1(a) is used in the implementation of high order MASH

DSMs and the model in Fig. 1(b), which is called the error feedback modulator, is used for calculations. The carry bit (the quantizer output in Fig. 1(b)) is determined by the following relationship:



(a) Digital Accumulator (b) The Accumulator Model

Fig. 1. The digital accumulator used in MASH DSMs (a) and its model (b)

$$c_n = \begin{cases} 0, & x_n + u_n < 2^N \\ 1, & x_n + u_n \geq 2^N \end{cases} \quad (1)$$

where N is the accumulator word length. If the digital word is normalized to unity, the quantizer input-output relationship is described by (2) and is shown in Fig. 2

$$y_n = \begin{cases} 0, & v_n < 1 \\ 1, & v_n \geq 1 \end{cases} \quad (2)$$

The ranges of the signals are:

$$x_n, u_n \in \{0, \frac{1}{2^N}, \dots, (1 - \frac{1}{2^N})\} \quad (3)$$

$$v_n \in \{0, \frac{1}{2^N}, \dots, 1, \dots, (2 - \frac{1}{2^{N-1}})\} \quad (4)$$

The nonlinear difference equations governing the structure shown in Fig. 1(b) can be summarized as:

$$v_n = x_n + u_n \quad \forall n \geq 0, \quad (5)$$

$$y_n = Q(v_n) = \begin{cases} 0 & v_n < 1 \\ 1 & v_n \geq 1, \end{cases} \quad (6)$$

$$u_n = \begin{cases} e_{n-1} & \forall n \geq 1 \\ u_0 & n = 0, \end{cases} \quad (7)$$

and the binary quantizer error is defined as:

$$e_n = v_n - y_n. \quad (8)$$

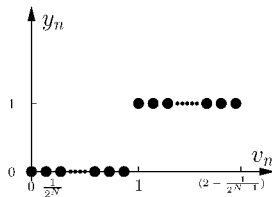


Fig. 2. Input-output relationship of the quantizer shown in Fig. 1(b)

With this definition for e_n , and considering (4) and (6), its values are in the range of:

$$e_n \in \{0, \frac{1}{2^N}, \dots, (1 - \frac{1}{2^N})\} \quad (9)$$

In (7), u_0 is the initial condition of the registers in Fig. 1. Using the above equations we find an expression for e_n versus x_n and u_0 . Starting with (8), we consider two cases:

Case 1: If $v_n \geq 1$, using (2) we can write:

$$e_n = v_n - 1 \quad \forall n \geq 0, \quad (10)$$

and considering (4) and (9) we can write:

$$e_n = \langle v_n \rangle \quad (11)$$

where $\langle x \rangle$ is the fractional part of x .

Case 2: If $v_n < 1$, using (2) we can write:

$$e_n = v_n - 0 \quad \forall n \geq 0 \quad (12)$$

and considering (4) and (9), we can write:

$$e_n = \langle v_n \rangle \quad (13)$$

Therefore (13) stands for all v_n in its range. Using (5), we have:

$$e_n = \langle x_n + u_n \rangle \quad \forall n \geq 0. \quad (14)$$

Rewriting the above equation and using (7), we have:

$$e_n = \langle x_n + e_{n-1} \rangle \quad \forall n > 0. \quad (15)$$

Expanding the above equation with its indices, we have:

$$e_0 = \langle x_0 + u_0 \rangle, \quad (16)$$

$$e_1 = \langle x_1 + e_0 \rangle, \quad (17)$$

and for index n we have:

$$e_n = \langle x_n + e_{n-1} \rangle. \quad (18)$$

If we replace the value of e from its values in previous equations, we can write e_n as

$$e_n = \langle x_n + \langle x_{n-1} + \langle x_{n-2} \dots + \langle x_0 + u_0 \rangle \rangle \rangle. \quad (19)$$

Using the property of fractional operators [6] that

$$\langle \langle a \rangle + \langle b \rangle \rangle = \langle \langle a \rangle + b \rangle = \langle a + \langle b \rangle \rangle = \langle a + b \rangle \quad (20)$$

we can rewrite (19) as:

$$e_n = \langle x_n + x_{n-1} + x_{n-2} \dots + x_0 + u_0 \rangle \quad (21)$$

$$= \langle u_0 + \sum_{k=0}^n x_k \rangle. \quad (22)$$

In the special case of a constant dc input X , we have:

$$e_n = \langle u_0 + (n+1)X \rangle \quad \forall n \geq 0. \quad (23)$$

If there is a period of N_1 for e_n , then we should have:

$$e_n = e_{n+N_1} = \langle u_0 + (n+N_1+1)X \rangle = \langle u_0 + (n+1)X + N_1X \rangle \quad (24)$$

and therefore the term $N_1 X$ should be an integer. From (24) it is obvious that the sequence length of the first order modulator does not depend on initial state u_0 . We know that X is a normalized rational dc input and we can write it as:

$$X = d_1/2^N = d_1/M. \quad (25)$$

The solutions for N_1 so that $N_1 X$ is an integer are summarized in Table II.

TABLE II

THE SEQUENCE LENGTH OF THE FIRST ORDER ERROR FEEDBACK MODULATOR RELATED TO ITS INPUT DIGITAL WORD

Digital Input, d_1	$k_1 2^k$, where k_1 is odd and $0 \leq k < N$
Sequence Length	$M/2^k$

The maximum sequence length (M) is only achieved when the input is an odd number.

III. HIGHER ORDER MODULATORS

Higher order MASH DSMs are composed of the simple structure investigated in the previous section (EFM blocks in Fig. 3). The general structure of an l^{th} order MASH DSM is shown in Fig. 3. The structure consists of several cascaded first order error feedback modulators (EFM). The input to each stage is the quantizer error of the previous stage. The noise cancellation network is used to eliminate the quantizer errors of the stages. The quantizer error of the last stage appears directly in the output spectrum and therefore we find a closed form expression for the binary quantizer error of the last stage using the method presented in the previous section. We use (23) and (22) for the first stage and the other stages respectively.

$$e1_n = \langle u_1 + (n+1)X \rangle \quad \forall n \geq 0, \quad (26)$$

$$e2_n = \langle u_2 + \sum_{k=0}^n e2_k \rangle, \quad (27)$$

and for the last stage we have:

$$el_n = \langle u_l + \sum_{k=0}^n e(l-1)_k \rangle. \quad (28)$$

Using the expression of $e(l-1)_k$ from the previous indices, we can write:

$$\begin{aligned} el_n = & \langle u_l + \sum_{k_{l-1}=0}^n \langle u_{l-1} + \sum_{k_{l-2}=0}^{k_{l-1}} \langle u_{l-2} + \dots \\ & + \sum_{k_1=0}^{k_2} \langle u_1 + (k_1+1)X \rangle \rangle \rangle \rangle \end{aligned} \quad (29)$$

and using (20) we can write:

$$el_n = \langle u_l + \sum_{k_{l-1}=0}^n u_{l-1} + \sum_{k_{l-2}=0}^{k_{l-1}} u_{l-2} + \dots + \sum_{k_1=0}^{k_2} u_1 + (k_1+1)X \rangle \quad (30)$$

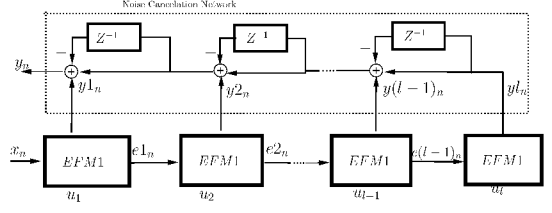


Fig. 3. Block diagram of the digital MASH DSM. $EFM1$ Blocks are the structure shown in Fig 1(b).

A. Second Order Modulators ($l = 2$)

We find the sequence length of the second order modulator using (30). After expanding (30), we have:

$$e2_n = \langle u_2 + \sum_{k=0}^n u_1 + (k+1)X \rangle. \quad (31)$$

If there is a period of N_2 , we should have:

$$e2_n = e2_{n+N_2} = \langle u_2 + \sum_{k=0}^{n+N_2} u_1 + (k+1)X \rangle \quad (32)$$

$$= \langle u_2 + \sum_{k=0}^n (u_1 + (k+1)X) + \sum_{k=n+1}^{n+N_2} (u_1 + (k+1)X) \rangle \quad (33)$$

The last term should be an integer; thus, we have:

$$\sum_{k=n+1}^{n+N_2} u_1 + (k+1)X = \text{an integer} \quad \forall n \geq 0 \quad (34)$$

Examining (34) at $n = 0$ reduces it to:

$$\sum_{k=1}^{N_2} u_1 + (k+1)X. \quad (35)$$

Further expansion of this expression, yields the condition

$$N_2 u_1 + \frac{N_2(N_2+3)}{2} X = \text{an integer}, \quad (36)$$

where u_1 is the initial condition on the first stage and X is the normalized digital dc input. We replace u_1 and X with their corresponding rational representations:

$$N_2 \frac{d_2}{M} + \frac{N_2(N_2+3)}{2} \frac{d_1}{M} = \text{an integer} \quad (37)$$

which d_2 is the digital initial condition on the first accumulator, d_1 is the digital input word and M is equal to 2^N . We examine the periodicity of (37) with an odd initial condition. In case of odd digital inputs, considering (37) the sequence length, N_2 , is $2M$. In case of even digital inputs we consider two cases:

Case 1: If the input is in the form of, $2k$; when k is odd; we can write (37) as:

$$\frac{N_2}{M} (d_2 + (N_2+3)k). \quad (38)$$

The expression $(d_2 + (N_2 + 3)k)$ is even (k and d_2 are odd) therefore, the solution for N_2 so that (38) is an integer, is $\frac{M}{2}$. Case 2: If the input is in the form of $4k$, we can write 37 as:

$$\frac{N_2}{M}(d_2 + (N_2 + 3)2k), \quad (39)$$

Since the term $(d_2 + (N_2 + 3)2k)$ is odd, the solution for N_2 so that (39) is an integer, is M . Table III summarizes the resulting sequence lengths of the second order modulator.

TABLE III

THE SEQUENCE LENGTHS OF THE SECOND ORDER MASH DSM RELATED TO ITS DIGITAL INPUT WORDS AND INITIAL CONDITION ON THE FIRST STAGE

d_2	d_1	N_2
odd	odd	$2M$
odd	$4k, k$ is integer	M
odd	$2k, k$ is odd	$\frac{M}{2}$

The maximum sequence length $2M$ is achieved when the input and the initial condition of the first stage are odd.

B. Third Order Modulators ($l = 3$)

We expand (30) for the third order modulator as follows:

$$e3_n = \langle u_3 + \sum_{k_2=0}^n (u_2 + \sum_{k_1=0}^{k_2} (u_1 + (k_1 + 1)X)) \rangle. \quad (40)$$

If there is period of N_3 , then the following condition should hold:

$$e3_0 = e3_{N_3} \quad (41)$$

After expanding (40), applying the following mathematical identities

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (42)$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad (43)$$

and following the same procedure used for the second order case, we find (44) that the sum (44) should be an integer.

$$N_3 u_2 + \frac{N_3(N_3 + 3)}{2} u_1 + \frac{X}{2} \left(\frac{N_3(N_3 + 3)(N_3 + 5)}{3} + 2N_3 \right) \quad (44)$$

where u_1 and u_2 are the normalized initial conditions on the first and the second stages. Equation (44) shows that the sequence length does not depend on the initial condition on the last stage. We use normalized rational representation for u_1 and X and assume that u_2 is zero.

$$\frac{d_2}{M} \frac{N_3(N_3 + 3)}{2} + \frac{d_1}{M} N_3 + \frac{d_1}{2M} \left(\frac{N_3(N_3 + 3)(N_3 + 5)}{3} \right) \quad (45)$$

$$= a + b + c, \quad (46)$$

where d_1 and d_2 are the digital input and odd initial condition on the first stage, respectively. To find N_3 so that (45) is an integer, we define three terms as a, b and c , respectively. Since d_2 is odd, the minimum value for N_3 so that a is an integer is $2M$. The minimum value of N_3 so that b is an integer when d_1 is even is equal to $\frac{M}{2}$ and, when it is odd, is equal to M . In case of $\frac{M}{2}$, the first term a is not integer; therefore the minimum value for N_3 in order for b to be an integer is M . Next we want to prove that for the third term c , the minimum value of N_3 is M when d_1 is even, and when it is odd, is equal to $2M$. We can write c as

$$\frac{N_3(N_3 + 1)(N_3 + 5)}{6} \frac{d_1}{M} = \text{an integer} \quad (47)$$

Equivalently,

$$N_3(N_3 + 1)(N_3 + 5)d_1 = 3k2^{N+1} = 3k(2M). \quad (48)$$

If d_1 is odd, then $d_1(N_3 + 1)(N_3 + 5)$ is odd (N_3 is even required by the second term b) and we can write:

$$N_3 = k2^{N+1}. \quad (49)$$

Thus the minimum value of N_3 is 2^{N+1} ($k = 1$) and when d_1 is even, N_3 is half of this value and is equal to $2^{N+1} (= M)$. Therefore, the smallest common multiplier among $2M$ for a , M for b and M or $2M$ for c is $2M$ for all inputs, d_1 .

IV. CONCLUSION

In this paper we demonstrated mathematically that, for an odd initial condition on the first accumulator of the second and third order modulators described in [7], the sequence length is determined from Table I and also for the first order modulator the sequence length is correctly determined from Table II, where $M = 2^N$. We found that the initial condition on the last stage has no effect on the periodicity of the modulator.

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